NatV conditional sequence modeling

Study design and information available to the categorical model

A. THREE INITIALIZATIONS, TWO MODEL ARMS

Random task weights
No auxiliary training

NatV train fragments
Matched self pretraining

External fragments
Screened homologous domains

QVAE + deterministic
Three pipeline replicates

QVAE + deterministic
Three pipeline replicates

QVAE + deterministic
Three pipeline replicates

B. QVAE: VISIBLE CONTEXT AND FULL-SEQUENCE POSTERIOR

Visible-context features
Masked ESM-2 + context encoder

Prior p(z | c, M)
16 categorical factors; 8 states

z ~ p

Shared FiLM decoder
20-amino-acid probabilities

Full-sequence features
Frozen ESM-2, unmasked

Posterior q(z | x, c, M)
Training / importance scoring

z ~ q

Initialization or visible-context path

Posterior path: not used for generation

All model arms use frozen ESM-2. QVAE training adds KL(q || p) to the reconstruction objective.